

LAC JOINT PROTEC COMPLEX

FILM-COATED TABLET

PRODUCT DESCRIPTION:

A white color film coated elliptical tablet.

COMPOSITION:

Each tablet contains: Glucosamine HCl 500 mg
Chondroitin Sulfate 400 mg
Methylsulfonylmethane (MSM) 300 mg

THERAPEUTIC PROPERTIES:

The actions of glucosamine/chondroitin/msm have yet to be clarified. It may play a role in the promotion and maintenance of the structure and function of cartilage in the joints of the body.

Glucosamine/Chondroitin/MSM may also have anti-inflammatory properties.

PHARMACODYNAMICS:

Until the specific actions of glucosamine are determined, the mechanism of action in relieving arthritic pain and in repair of cartilage is a matter of speculation. However, we do know a great deal about the biochemistry of the molecules in which glucosamine is found. Biochemically, glucosamine is involved in glycoprotein metabolism. Glycoproteins, known as proteoglycans, form the ground substance in the extra-cellular matrix of connective tissue. Proteoglycans are polyanionic substances of high-molecular weight and contain many different types of heteropolysaccharide side-chains covalently linked to a polypeptide-chain backbone. These polysaccharides make up to 95% of the proteoglycan structure. In fact, chemically, proteoglycans resemble polysaccharides more than their do proteins.

The polysaccharide groups in proteoglycans are called glycosaminoglycans or GAGs. GAGs include hyaluronic acid, chondroitin sulfate, dermatan sulfate, keratan sulfate, heparin and heparan sulfate. All of the GAGs contain derivatives of glucosamine or galactosamine.

Glucosamine derivatives are found in hyaluronic acid, keratan sulfate and heparan sulfate. Chondroitin sulfate contains derivatives of galactosamine.

The glucosamine-containing glycosaminoglycan hyaluronic acid is vital for the function of articular cartilage. GAG chains are fundamental components of aggrecan found in articular cartilage. Aggrecan confers upon articular cartilage shock-absorbing properties. It does this by providing cartilage with a swelling pressure that is restrained by the tensile forces of collagen fibers. This balance confers upon articular cartilage the deformable resilience vital to its function.

In the early stages of degenerative joint disease, aggrecan biosynthesis is increased. However, in later stages, aggrecan synthesis is decreased, leading eventually to the loss of cartilage resiliency and to most of the symptoms that accompany osteoarthritis.

During the progression of osteoarthritis, exogenous glucosamine may have a beneficial role. It is known that, in vitro, chondrocytes do synthesize more aggrecan when the culture medium is supplemented with glucosamine. N-acetylglucosamine is found to be less effective in these in vitro studies. Glucosamine has also been found to have antioxidant activity and to be beneficial in animal models of experimental arthritis.

The counter anion of the glucosamine salt (i.e. chloride or sulfate) is unlikely to play any role in the action or pharmacokinetics of glucosamine. Further, the sulfate in glucosamine sulfate supplements should not be confused with the glucosamine sulfate found in such GAGs as keratan sulfate and heparan sulfate. In the case of the supplement, sulfate is the anion of the salt. In the case of the above GAGs, sulfate is present as an ester. Also, there is no glucosamine sulfate in chondroitin sulfate.

Until the specific actions of chondroitin sulfate are determined, the mechanism of action is a matter of speculation. However, much is known about the biochemistry and physiology of chondroitin sulfate and similar molecules. Glycoproteins known as proteoglycans form the ground substance in the extracellular matrix of connective tissue. Proteoglycans are polyanionic substances of high molecular weight and contain heteropolysaccharide-side-chains covalently linked to a polypeptide-chain backbone. The polysaccharides, which include chondroitin sulfate and hyaluronic acid, make up as much as 95% of the proteoglycan structure.

The polysaccharides in proteoglycans are called glycosaminoglycans or GAGs. Chondroitin sulfate and hyaluronic acid are vital for the structure and function of articular cartilage. Chondroitin sulfate and hyaluronic acid are fundamental components of aggrecan found in articular cartilage. Aggrecan confers upon articular cartilage shock-absorbing properties. It does this by providing cartilage with a swelling pressure that is restrained by the tensile force of collagen fibers. This balance confers upon articular cartilage the deformable resilience vital to its function. Hyaluronic acid, which is also found in synovial fluid, has lubricating properties for the joint.

In the progression of degenerative joint disease or osteoarthritis, aggrecan synthesis is decreased, leading to the loss of cartilage resiliency and the pain and other symptoms that accompany osteoarthritis.

Methylsulfonylmethane (MSM) has been proven to have anti-inflammatory and antioxidant mechanisms in an *in vitro* study in which human neutrophils were artificially stimulated to produce oxidative compounds, including hydrogen peroxide, superoxide, and hypochlorous acid. After cell lines were treated with either Dimethyl sulfoxide (DMSO) or Dimethyl Sulfone (DMSO₂), these free radical by-products were decreased.

It has been suggested that polar solvents such as MSM and DMSO have a chemo preventive mechanism that affects the interaction of tumour cells with the host immune response. Researchers examined DNA of DMSO-treated cells and found these polar solvents create DMSO-induced nicks found in the DNA of folded genomes, suggesting these nicks may cause an untwisting of DNA, with resultant transcription of additional genes. Additional chemoprevention studies suggest MSM has no effect on cyclooxygenase (COX) activity or PGE₂ activity, and operates on a COX-independent pathway in inducing differentiation.

An early study of DMSO and DMSO₂ demonstrates these agents reduce the binding, uptake, and degradation of low-density lipoproteins by cultured fibroblasts. A similar study showed DMSO and DMSO₂ caused a dose-dependent suppression of growth and proliferation of cultured aortic smooth muscle and endothelial cells *in vitro*, the more substantial effect occurring in smooth muscle cells. In addition, DMSO₂ was a more potent inhibitor of cell growth than DMSO and its effects were more irreversible than the effects of DMSO.

PHARMACOKINETICS:

Pharmacokinetics of glucosamine are derived primarily from animal studies. About 90% of glucosamine administered orally as a glucosamine salt gets absorbed from the small intestine, and from there it is transported via the portal circulation to the liver. It appears that a significant fraction of the ingested glucosamine is catabolized by first-pass metabolism in the liver. Free glucosamine is not detected in the serum after oral intake, and it is not presently known how much of an ingested dose is taken up in the joints in humans. Some uptake in the articular cartilage is seen in animal studies.

Earlier studies using high-molecular-weight chondroitin sulfate, concluded that there was no significant absorption of this high-molecular-weight version of chondroitin sulfate. More recent studies demonstrate that there is probably significant absorption of low-molecular-weight chondroitin sulfate. Absorption appears to occur from the stomach and small intestine. There is also an indication that some chondroitin sulfate, after absorption, does enter the joint space.

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It is of interest to note that a molecule similar in many respects to chondroitin sulfate, pertosan polysulfate, approved for the treatment of interstitial cystitis, is given orally and is absorbed to some extent.

A study with Rhesus monkeys on metabolism and excretion of DMSO found the primary metabolite DMSO₂ became detectable in serum approximately two hours after ingestion of DMSO. With continued DMSO ingestion, DMSO₂ maintained a steady concentration in the serum. When DMSO was stopped after 14 days, the mean DMSO₂ concentration declined slowly over the subsequent 96 hours, and only trace amounts were detectable after five days. The decline in serum DMSO₂ was linear, and its half-life appeared to be about 38 hours. The authors observed that absorption in these animals was similar to humans, but elimination was quicker in the monkeys. DMSO₂ has been shown to persist in the blood up to five times longer than DMSO.

ROUTE OF ADMINISTRATION:

Oral

EFFECT ON ABILITY TO DRIVE & USE MACHINE:

No Applicable

INDICATIONS:

Adjunctive therapy in osteoarthritis.

ADVERSE/SIDE EFFECTS:

Cardiovascular:

Peripheral oedema, tachycardia was reported in a few patients following larger clinical trials investigating oral administration in osteoarthritis. Causal relationship has not been established.

Central Nervous System:

Drowsiness, headache, insomnia have been observed rarely during therapy (less than 1%)

Gastrointestinal:

Nausea, vomiting, diarrhea, dyspepsia or epigastric pain, constipation, heartburn and anorexia have been described rarely during oral therapy with glucosamine

Skin:

Skin reactions such as erythema and pruritus have been reported with therapeutic administration of glucosamine.

CONTRA-INDICATION:

Hypersensitivities to glucosamine and patients who are allergic to shellfish.

WARNING AND PRECAUTIONS:

Glucosamine may increase insulin resistance. Glucosamine increases insulin resistance in normal and experimentally diabetic animals. In these animals, intravenous glucosamine significantly decreases the rate of glucose uptake in skeletal muscle. In animals given oral glucosamine, this is not observed.

Those with type 2 diabetes and those who are overweight and have problems with glucose tolerance should have their blood sugars carefully monitored if they use glucosamine supplements.

To include anti-inflammatory drug (in case of severe pain) during first few days of treatment. Glucosamine (and chondroitin/MSM) are not meant to replace NSAIDs in cases of intense pain as this is a causal therapy.

Glucosamine HCL is derived from seafood and Chondroitin Sulfate from bovine source.

PREGNANCY AND LACTATION:

Because of insufficient safety data, children, pregnant women and nursing mothers should avoid using glucosamine.

Pregnancy and lactation should be use only under medical control.

Administration during the first three months of pregnancy must be avoided.

DRUG INTERACTION:

Glucosamine may increase insulin resistance and consequently affect glucose tolerance. Diabetics who, under medical advisement, decide to use glucosamine supplements will need to monitor their blood glucose and may need to adjust the doses of the medications they take to control blood glucose. This needs to be done under medical supervision. No other drug, nutritional supplement, food or herb interaction is known.

There is a possibility that chondroitin sulphate may have anti-thrombotic activity. Those taking warfarin and those with hemophilia should exercise caution in its use. Chondroitin should not be used concomitantly with chitosan.

RECOMMENDED DOSAGE:

Light to Moderate: 1 tablet twice daily at least 6 weeks, before meals (or according to medical prescription)
Severe Osteoarthritis Symptoms:

Initial therapy: 1 tablet, 3 times daily during a period of at least 12 weeks.
Follow-up therapy: Maintenance therapy should be followed for 3-4 months (or according to medical prescription) by administration of 1 tablet taken twice daily.

SYMPTOMS AND TREATMENT FOR OVERDOSAGE AND ANTIDOTE:

In the event of an acute overdose, the patient should be observed closely, and symptomatic treatment should be instituted.

SHELF LIFE:

Three years from manufacturing date.

PACKING/PACK SIZES:

In a bottle of 90 tablets.

STORAGE CONDITIONS:

Store in a dry place below 30°C in a tightly closed container.

Keep out of reach of children.

Jauhkan daripada kanak-kanak.

Further information concerning this drug can be obtained from your family physician/local general practitioner/pharmacist.

Reg. No: MAL23086024XC

Product Registration Holder:

LAC Global Brands (Malaysia) Sdn. Bhd. (545007-W),
No 8, Jalan 6/91, Taman Shamelin Perkasa,
Batu 3½, Jalan Cheras, 56100 Kuala Lumpur.

Manufactured for LAC Global Pte. Ltd. (a subsidiary of V3 Brands) by Y.S.P Industries (M) Sdn. Bhd,
Lot 3, 5, 7, 12 & 14, Jalan P/7, Section 13,
Kawasan Perindustrian Bandar Baru Bangi,
43300 Kajang, Selangor.

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